

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

**INFR11157 NATURAL LANGUAGE UNDERSTANDING,
GENERATION, AND MACHINE TRANSLATION (LEVEL 11) PG
AND INFR11225 UG**

Tuesday 6th May 2025

13:00 to 15:00

INSTRUCTIONS TO CANDIDATES

1. Note that **ALL QUESTIONS ARE COMPULSORY.**
2. **DIFFERENT QUESTIONS MAY HAVE DIFFERENT NUMBERS OF TOTAL MARKS.** Take note of this in allocating time to questions.
3. This is a **NOTES NOT PERMITTED, CALCULATORS NOT PERMITTED** examination.

Notes and other written or printed material **MAY NOT BE CONSULTED** during the examination.

CALCULATORS MAY NOT BE USED IN THIS EXAMINATION.

Year 4 Courses

Convener: M.Cryan

External Examiners: D.Parker, A.Pocklington, R.Mullins

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. Laura was asked to construct an algorithm that would “watermark” large language model text, so text generated by an LLM can later be automatically identified by another verification algorithm. Laura thought about the following algorithm. Given a language model, the algorithm works by apriori dividing its vocabulary set into two subsets of *roughly equal* probability (0.5) by estimating a word probability based on counts from a large corpus. The two sets are: one set of “green” tokens and one set of “red” tokens. Then, given a prompt, the LLM is constrained to only generate text using the green tokens. Following that modification to the LLM, Laura’s idea was to test, for any text, whether it uses green tokens only. If it does, then we can classify it as generated by the LLM.

- (a) Explain in two sentences one major disadvantage of this algorithm. Will it work in practice? [3 marks]
- (b) Laura further decided to improve her algorithm, and rather than dividing the vocabulary set apriori, her algorithm now divides the vocabulary set into a set of green and red tokens per *prompt*, randomly as in her original algorithm, independently of the prompt. What is the major issue with this new modification? [3 marks]
- (c) Suggest an alternative that you expect would work well, and overcome both the first disadvantage and the second disadvantage in the above subquestions (a) and (b). You can base your answer on an existing algorithm described in class. [5 marks]
- (d) Note briefly one disadvantage of the algorithm you proposed. [4 marks]

2. Alex is working for an international firm that wants to provide a reliable open-domain question answering (QA) system. They decide to frame the task as a retrieval augmented generation (RAG) system based on Wikipedia. Please answer the following questions being as concise as possible. Please do not write more than one or two sentences per mark assigned.

- (a) Explain one advantage and two disadvantages of using RAG over directly querying a large language model trained on Wikipedia. [6 marks]
- (b) The QA product took only the English wikipedia as its data source, and to serve their Arabic speaking customers they translated queries from Arabic into English, and then they translated answers from English into Arabic. Please describe and explain any problems that they might have encountered with the quality of the answers returned using this approach. Please comment on the ethical risks with this approach. [2 marks]
- (c) Please suggest two ways Alex could incorporate the non-English Wikipedia articles into the QA system and the advantages and disadvantages of these two methods. [4 marks]
- (d) Using an example, describe what problems the system would encounter when searching for answers to questions where information is likely to be very different between English Wikipedia and Arabic Wikipedia. Think about how a system could handle these discrepancies in a way that is useful for a user. [3 marks]

3. The following is a set of multiple choice questions. For each question, choose the answer that is the most correct. Each question is worth 2 marks. [20 marks]

- (a) Which of the following statement is incorrect? Mark D if none of them are incorrect.
- A. The sum of two power laws is a power law.
 - B. Scaling laws for LLMs are often based on power laws.
 - C. Let $\alpha \geq 0$ be fixed. To calculate $f(\alpha x)$ for a power law $f(x)$, we need to multiply $f(x)$ by a constant.
 - D. None of the statements above is incorrect.
- (b) Compared to ROUGE, BERTScore improves the automatic evaluation of summaries because:
- A. It is more efficient.
 - B. It removes the need to rely on strict lexical overlap.
 - C. It calculates an F_1 measure, which ROUGE cannot do.
 - D. All of the statements above are correct.
- (c) A disadvantage of greedily choosing the maximal probability word at each step of decoding from an LLM is:
- A. It is less memory efficient than beam decoding.
 - B. We cannot backtrack in the case of choosing low probability paths.
 - C. We should always select a word that is in the mid-range of probabilities, as in locally typical sampling.
 - D. None of the statements above is correct.
- (d) Locally typical sampling works by:
- A. Sampling words at random from the next-token distribution.
 - B. Increasing the probability of words by a factor proportional to their entropy.
 - C. Choosing words to output that are closest to the entropy of the next-token distribution.
 - D. Removing words which have very low probability.
- (e) With neural parsing, an advantage of the Transformer parsing model we learned about compared to the LSTM parsing model is:
- A. It can be used with backpropagation.
 - B. It does not use a dynamic programming algorithm for parsing.
 - C. It achieves good results without the need for additional parsed data.
 - D. Statements A and C are correct.

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- (f) Which component of the LSTM language model is responsible for fixing the vanishing gradient problem:
- A. The activation functions
 - B. The linear memory cell
 - C. The embedding layer
 - D. A. and B.
- (g) In the Word Embeddings Association Test (WEAT) we have embeddings for two groups (male and female names), and two stereotypes (science terms and family terms). We are measuring:
- A. If female names are closer to family terms than science terms.
 - B. If male names are closer to family terms than science terms.
 - C. If male names are closer to science terms than to family terms, when compared to female names.
 - D. A. and C.
- (h) Measuring agreement among annotators using Cohen's Kappa statistic (where $P(A)$ is the proportion of times the annotator agree, and $P(E)$ is the proportion of times that annotators would agree by chance) we need to divide by:
- A. $1 - P(A)$
 - B. $P(A)$
 - C. $P(E)$
 - D. $1 - P(E)$
- (i) Comparing the translation output of an encoder-decoder translation model trained mainly on parallel data, to a multilingual language model, would you expect that the LLM:
- A. is better able to translate toxic content.
 - B. is less literal .
 - C. is less likely to produce non-standard orthography for things like article titles with no full stop at the end.
 - D. B. and C.

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- (j) Which of the following statements about the Natural Questions (NQ) dataset is TRUE?:
- A. The dataset consists only of manually written questions without any connection to real-world queries.
 - B. It includes questions naturally asked by users on Google Search, along with corresponding Wikipedia passages.
 - C. Natural Questions only provides binary yes/no answers for all questions.
 - D. The dataset exclusively contains short factoid-style answers with no long-form responses.